

Geometric & Topological Methods in Atmospheric Dynamics

Reference Card — UAH Graduate Course

I. Differential Geometry

- $\mathcal{M}$: physical spatial manifold (Earth/domain)
- $T\mathcal{M}$: vector fields u ; $T^*\mathcal{M}$: covectors/1-forms v
- Forms: Ω^0 scalars, Ω^1 circulation, Ω^2 area density

Operator dictionary

Op	Meaning
d	exterior derivative
$\star$	Hodge star (metric dependent)
$\wedge$	wedge product
ι_u	contraction with u
$\mathcal{L}_u$	Lie derivative (advection)

II. Exterior Calculus

- Graded-commutative: $\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha$, $\alpha \in \Omega^p$, $\beta \in \Omega^q$
- General Hodge identity: $\star^2 = (-1)^{k(n-k)}$ on $\Omega^k(\mathcal{M})$
- Cartan formula: $\mathcal{L}_u = d\iota_u + \iota_u d$
- Stokes: $\int_{\partial M} \omega = \int_M d\omega$

Core identities

$$\begin{aligned} d^2 &= 0 & \mathcal{L}_u &= d\iota_u + \iota_u d & \int_{\partial M} \omega &= \int_M d\omega \\ \delta &= \star d \star & \Delta &= d\delta + \delta d & \Pi d &\equiv d\Pi \end{aligned}$$

III. Geometric Fluid Dynamics

- Incompressible Euler (vector invariant): $\partial_t v + \iota_u(dv) + dB = 0$, $B = \frac{1}{2} \iota_u v + p + \Phi$
- Incompressibility: $d(\star v) = 0$ (equiv. $\nabla \cdot u = 0$)
- Vorticity transport: $\partial_t(\star \zeta) + \mathcal{L}_u(\star \zeta) = 0$
- PV: $q = (\zeta + f)/h$, material conservation $(\partial_t + \mathcal{L}_u)q = 0$

Invariants

$$E = \frac{1}{2} \int_M v \wedge \star v; \quad \frac{d}{dt} \oint_{c(t)} v = 0; \quad Z = \frac{1}{2} \int_M \zeta^2 dA.$$

IV. DEC / FEEC

- Principle: topology exact, geometry approximate via $\star$
- Primal/dual complexes: Delaunay/Voronoi pairing
- Exact incidence algebra: discrete $d^2 = 0$
- Compatible sequence: $V_0 \xrightarrow{\nabla^\perp} V_1 \xrightarrow{\nabla \cdot} V_2$ (scalar/flux/cell)
- Commuting projection: $\Pi_{k+1} d = d \Pi_k$ (suppresses spurious modes)
- Skew advection: $\langle v, \mathcal{L}_u v \rangle_1 = 0$ (energy consistency)

V. Dynamical Systems & GSPT

Notation: M_0, M_ϵ denote phase-space critical/slow manifolds (not spatial $\mathcal{M}$).

Local bifurcations: saddle-node $\dot{x} = \mu - x^2$; pitchfork $\dot{x} = \mu x - x^3$; transcritical $\dot{x} = \mu x - x^2$; Hopf $\dot{z} = (\mu + i\omega)z - z|z|^2$.

Fast-slow template

Fast: $\dot{x} = f(x, y, \epsilon)$ Slow: $\epsilon \dot{y} = g(x, y, \epsilon)$
Critical manifold: $M_0 = \{g(x, y, 0) = 0\}$ Normal hyperbolicity: $\Re \lambda_{\text{fast}} \neq 0$

Fenichel: $M_0 \rightarrow M_\epsilon$ Loss NH: fold region $\rightarrow$ canards.

Catastrophes: fold (codim-1), cusp (codim-2), swallowtail (codim-3).

VI. LCS & TDA

Space split: LCS live on physical $\mathcal{M}$; M_ϵ lives in state space.

LCS chain: flow map $F_{t_0}^t$; $C = (DF)^\top DF$; $\sigma = \frac{1}{|t - t_0|} \ln \sqrt{\lambda_{\max}(C)}$.

- Repelling LCS: strongest forward stretching lines
- Attracting LCS: strongest forward squeezing lines

- Elliptic LCS: closed null-geodesics (objective coherent vortices)
- Parabolic LCS: null-geodesic shear lines (jet cores)
- TDA pipeline:** cloud $\rightarrow$ filtration $\rightarrow H_k \rightarrow$ barcode/diagram.
- β_0 : components; β_1 : loops; β_2 : voids
- Persistence point (b_i, d_i) ; lifetime $d_i - b_i$ separates signal/noise

VII. Atmospheric Applications

Concept	Geometry
MOST	1D manifold
SCM/regime branch	slow manifold
Ri_c	fold / loss of NH
SBL transition	canard-type jump
Turbulence collapse	NH breakdown
Diurnal cycle	hysteresis loop

Geometry $\leftrightarrow$ Physics

Geometry	Atmosphere
0-form	scalar field
1-form	circulation
2-form	mass/area density
d	grad/curl/div structure
$\star$	metric map
$\mathcal{L}_u$	advection
critical manifold	equilibrium states
fold	tipping point
persistence	coherent structure

Dimensionless numbers: $Ri = \frac{(g/\theta_v)\partial_z \theta_v}{(\partial_z u)^2 + (\partial_z v)^2}$, $Ro = \frac{U}{fL}$, $Fr = \frac{U}{NH}$.